

API Security Risk Analysis Report

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Introduction

APIs enable communication between applications. Poor API security can lead to unauthorized access, data breaches, and service disruption.

Objective

Identify common API security risks and recommend mitigation measures.

Tools Used

Postman, Browser Developer Tools, Sample REST API.

Findings

1. Missing Authentication (High)
2. Excessive Data Exposure (Medium)
3. Missing Rate Limiting (Medium)
4. Weak Input Validation (High)

Business Impact

Unauthorized access, information disclosure, denial-of-service, and data manipulation.

Recommendations

Use JWT/OAuth authentication, validate inputs, implement rate limiting, minimize returned data, monitor API logs, and enforce HTTPS.

Conclusion

Implementing secure authentication, validation, and monitoring significantly improves API security.